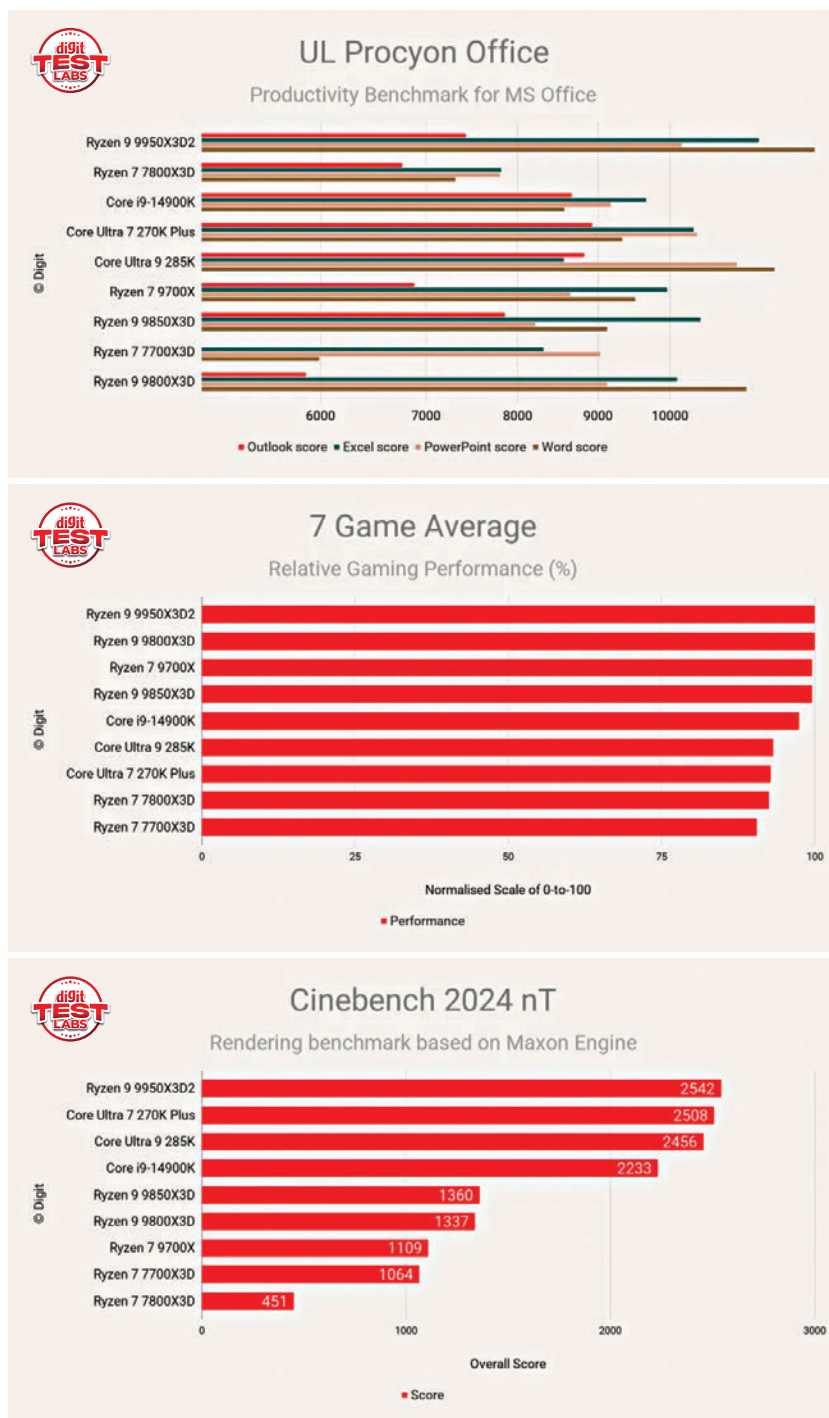




we have to rely on popular AAA video games to assess how well the processor deals with contemporary game engines. We revise the video games once every two years based on what's most popular as per Steam and we also maintain a mix between different game engines.

Thermals and power

On paper, the Ryzen 7 7700X3D has a 120 W TDP and a 162 W default PPT limit. AMD lists an 89°C maximum junction temperature and recommends a 240-to-280 mm liquid cooler or an equivalent high-performance air cooler.

These limits match those of the Ryzen 7 7800X3D. The lower frequency ceiling may allow the 7700X3D to consume less power than the 7800X3D in some lightly threaded and gaming workloads.

VERDICT

The Ryzen 7 7700X3D is a focused response to a fairly specific buying question i.e. how much of the 7800X3D's gaming performance can be retained at a lower price? From the results, the answer is most of it. The processor was approximately 4 per cent slower than the 7800X3D in gaming whereas against the regular Ryzen 7 7700X, it produced an average gaming improvement of roughly 8-9 per cent and delivered gains exceeding 15 per cent in certain cache-sensitive titles. That makes the 7700X3D a substantial gaming upgrade over the 7700X. The compromise is equally clear. A maximum boost frequency of 4.5 GHz leaves the processor behind the 7700X in most productivity benchmarks. It also trails the 7800X3D outside gaming, despite sharing the same core and cache configuration.

A Ryzen 7 7800X3D retails for about Rs 42K in India and there's a Ryzen 5 7500X3D available at Rs 25K. A modest discount over the 7800X3D would not be enough, because the older processor is consistently faster and avoids some of the 7700X3D's productivity limitations. A meaningful price gap would make the new model an appealing entry point into eight-core 3D V-Cache gaming.

—Mithun Mohandas

SPECIFICATIONS

Socket: AM5 | Core Process: 5nm | Core Architecture: Zen 4 | Cores: 8 | Threads: 16 | Integrated Graphics: RDNA2 | IG Cores: 2 | IG Base Frequency: 400 MHz | Cache: L1: 64, L2: 1, L3: 96 | Max Memory Capacity: 128 | TDP: 120

CONTACT

Company: AMD | Telephone: NA | Email: support.amd.com | Website: <https://www.amd/>

Read more on Digit.in:
tinyurl.com/c8uk1x1